
Local ZZ Interactions Encode Plasma-Wave Self-Phase Modulation in a Qubit Chain

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Abstract

Nonlinear plasma-wave envelopes acquire an intensity-dependent phase, but the qubit-chain representation demonstrated for linear plasma propagation has so far been confined to one excitation. We add one fixed interaction, $H_{ZZ} = -(\epsilon J/4) \sum_j Z_j Z_{j+1}$, without modifying the chiral-hopping and staggered-mass Hamiltonian. An exact hard-core-boson identity followed by a spin-coherent, long-envelope limit gives a cubic nonlinear Schrödinger equation (NLSE) with $q = -2\epsilon J$ in the chain bulk. We test its parameter-free differential phase rate on a 12-qubit, full- 2^{12} statevector at four predeclared couplings spanning a factor of eight. The statevector/PDE relative errors are 0.024%, 0.017%, 0.009%, and 0.115%, all below a 15% acceptance bound; the unextended control has zero nonlinear phase rate. In a separate two-excitation witness at $\epsilon = 0.8$, 74.132% of the population moves to other multi-excitation basis configurations and the ZZ and H_0 distributions differ by 11.220%. At $\epsilon = 0$, paired augmented and unextended dispersion and propagation trajectories are bitwise identical. A separate nine-qubit Ramsey reproduction recovers all 18 frequencies at $\Delta/J \in \{0, 1/4\}$ within 4.20%, and packets propagate or reflect under the three standard mass profiles. Agreement is deliberately local in time: extending the phase fit past $Jt = 2.4$ yields a 21.4% error as entanglement invalidates the classical closure. These results establish a controlled nonlinear field coefficient, not merely an interacting spectrum, while sharply delimiting the regime in which a spin-1/2 chain represents a classical plasma envelope.

1 Introduction

The amplitude of a plasma wave can change its own phase velocity. In an envelope description this self-phase modulation (SPM) is the homogeneous solution of a nonlinear Schrödinger equation (NLSE); gradients of the same phase shift produce spectral broadening, and the competition with dispersion underlies modulational instability. The NLSE reduction of Langmuir-wave dynamics originates in the Zakharov system [Zakharov, 1972] and has been refined for collisionless and coupled electromagnetic–Langmuir waves [Dysthe and Pécseli, 1977, Gupta et al., 1981]. Nonlinear frequency shifts are sensitive to the fluid and kinetic regime [Liu and Dodin, 2015], and plasma SPM has been measured at relativistic intensity [Watts et al., 2002]. A useful simulator must therefore identify a physical nonlinear coefficient and validate it against a field equation, rather than showing only that interactions change a many-body spectrum.

Quantum approaches to plasma dynamics have largely targeted linearized equations: Vlasov evolution [Engel et al., 2019], collision–stream Maxwell solvers [Vahala et al., 2020], and block-encoded cold-plasma waves [Novikau et al., 2022]. Broader assessments emphasize both the promise and the output bottleneck of such encodings [Dodin and Startsev, 2021, Joseph et al., 2023]. A complementary local approach maps a first-order discretization of the plasma wave equation onto a spin chain. Its nine-qubit superconducting realization propagated and scattered single plasma-wave excitations and explicitly proposed longitudinal ZZ interactions as the route to nonlinear dynamics [Sundar et al.,

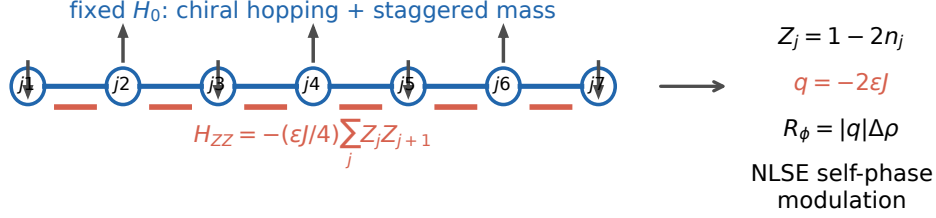


Figure 1: One fixed local extension. The blue terms are the unmodified chiral-hopping and staggered-mass chain; orange bonds are the attractive nearest-neighbor ZZ interaction. Rewriting $Z_j = 1 - 2n_j$ isolates a density–density attraction. In the coordination-two bulk it produces the envelope coefficient $q = -2\epsilon J$ and differential SPM rate $R_\phi = |q| \Delta \rho$.

2026]. Modal nonlinear wave mixing has separately been quantized on two superconducting qubits [Shi et al., 2024]; that method does not supply a spatially local envelope model.

Here we connect a single, nearest-neighbor ZZ extension to a specific plasma nonlinearity. The contribution has three parts. First, the exact density form of ZZ fixes the cubic NLSE coefficient without regression. Second, a 12-qubit dense statevector simulation tests the resulting intensity-differential phase rate at four couplings against a PDE solution computed before the quantum-model runs. Third, controls establish exact reduction to the linear chain and expose the time at which the mean-field correspondence fails. Figure 1 summarizes the construction.

2 Fixed spin model

For N open-boundary qubits we retain the linear Hamiltonian exactly as

$$H_0 = -\frac{J}{4} \sum_{j=1}^{N-1} (X_j Y_{j+1} - Y_j X_{j+1}) + \frac{1}{2} \sum_{j=1}^N \Delta_j (-1)^j Z_j, \quad (1)$$

and make the single model choice

$$H(\epsilon) = H_0 + H_{ZZ}(\epsilon), \quad H_{ZZ}(\epsilon) = -\frac{\epsilon J}{4} \sum_{j=1}^{N-1} Z_j Z_{j+1}. \quad (2)$$

Thus, under the convention $H_{ZZ} = \frac{1}{2} \sum_{ij} K_{ij} Z_i Z_j$, the only nonzero entries are $K_{j,j+1} = K_{j+1,j} = -\epsilon J/4$. The coupling is local, uniform, and attractive. It vanishes as an overall factor at $\epsilon = 0$; no on-site term in Eq. (1) is retuned.

Let $a_j = |0\rangle_j \langle 1|$ annihilate a hard-core excitation and $n_j = a_j^\dagger a_j = (1 - Z_j)/2$. Equations (1)–(2) are, up to constants,

$$H_0 = \frac{iJ}{2} \sum_j (a_j^\dagger a_{j+1} - a_{j+1}^\dagger a_j) - \sum_j \Delta_j (-1)^j n_j, \quad (3)$$

$$H_{ZZ} = -\epsilon J \sum_j n_j n_{j+1} + \frac{\epsilon J}{2} \sum_j (n_j + n_{j+1}) - \frac{\epsilon J}{4} (N - 1). \quad (4)$$

The first line is free after a Jordan–Wigner transformation in one dimension [Jordan and Wigner, 1928]; the quartic term in the second line is the new many-particle interaction. The remaining linear-density term is an intensity-independent carrier detuning. Our observable cancels it rather than adding a counterterm.

Nearest-neighbor density coupling is the shortest-range lattice representation of an adiabatic local refractive response: the wave intensity perturbs the neighboring plasma polarization, which shifts the carrier frequency. This captures the local cubic universality class of SPM. It is not a claim that one value of ϵ is a first-principles coefficient for every kinetic plasma; such a calibration depends on distribution function, temperature, and carrier mode [Liu and Dodin, 2015].

3 From the spin chain to the nonlinear envelope

3.1 Spin-coherent equation

We use a product spin-coherent state locally parameterized by occupation ρ_j and phase θ_j ,

$$|\chi_j\rangle = \sqrt{1-\rho_j}|0\rangle + e^{i\theta_j}\sqrt{\rho_j}|1\rangle, \quad \alpha_j \equiv \langle a_j \rangle = \sqrt{\rho_j(1-\rho_j)}e^{i\theta_j}. \quad (5)$$

This is the $S = 1/2$ coherent-state classical limit, closely related to the Holstein–Primakoff construction [Holstein and Primakoff, 1940]; coherent-state dynamics and its quantum corrections are reviewed in Zhang and Batista [2021]. The identity in Eq. (4) is exact; only the factorization of intersite correlators is semiclassical.

Using $[a_j, a_j^\dagger] = Z_j$ and factorizing distinct sites gives

$$\begin{aligned} i\dot{\alpha}_j &= \frac{iJ}{2}z_j(\alpha_{j+1} - \alpha_{j-1}) - \Delta_j(-1)^j\alpha_j + \frac{\epsilon J}{2}\alpha_j(z_{j-1} + z_{j+1}), \\ z_j &= 1 - 2\rho_j. \end{aligned} \quad (6)$$

The first term is the hard-core version of the discretized first-order wave operator. Its factor z_j generates a density–gradient correction even when $\epsilon = 0$; it vanishes identically for the homogeneous states used below. The last term is caused only by ZZ . In a uniform coordination-two bulk it becomes

$$i\dot{\alpha}|_{ZZ} = \epsilon J(1 - 2\rho)\alpha. \quad (7)$$

3.2 Long-envelope equation and measured coefficient

For a slowly varying carrier, projection of the linear spinor onto a band of Eq. (1) and expansion about k_0 produce group velocity v_g and curvature $D = \partial_k^2\omega(k_0)$. Moving with v_g , while expanding $\rho_{j\pm 1} = \rho(X) \pm a\partial_X\rho + \mathcal{O}(a^2)$, gives

$$i\partial_T\psi = -\frac{D}{2}\partial_X^2\psi + \underbrace{\epsilon J\psi}_{\text{carrier detuning}} \underbrace{-2\epsilon J|\psi|^2\psi}_{ZZ \text{ self phase}} + \mathcal{O}(a^2\partial_X^2\rho, \rho\partial_X\psi, \rho^2). \quad (8)$$

Here $|\psi|^2 = \rho + \mathcal{O}(\rho^2)$ and the phase of ψ equals that of α . Equation (8) has the standard cubic-NLSE form used for plasma-wave self-modulation [Zakharov, 1972, Dysthe and Pécseli, 1977, Asenjo, 2024]. The spin construction fixes

$$\mu_\epsilon = \epsilon J, \quad q(\epsilon) = -2\epsilon J. \quad (9)$$

Unlike a fitted phenomenological mapping, both coefficients follow from the coordination number and the normalization in Eq. (2).

For a homogeneous envelope the derivative terms vanish, so no assumption about D enters the present test. The solution is $\psi_\rho(T) = \sqrt{\rho}\exp[-i(\mu_\epsilon + q\rho)T]$. We cancel the Hartree detuning and all H_0 phase by comparing two densities:

$$\begin{aligned} R_\phi(\epsilon) &= |[\Omega_\epsilon(\rho_h) - \Omega_0(\rho_h)] - [\Omega_\epsilon(\rho_l) - \Omega_0(\rho_l)]| \\ &= 2\epsilon J(\rho_h - \rho_l). \end{aligned} \quad (10)$$

This differential angular frequency is SPM’s local nonlinear coefficient. Spatial chirp and modulational growth are consequences of the same q but are not inferred from the homogeneous experiment.

4 Numerical protocol

Frozen analysis. Before generating reference or statevector validation output, we fixed Eqs. (2) and (10), the four-point grid $\epsilon = (0.10, 0.20, 0.40, 0.80)$, densities $(\rho_l, \rho_h) = (0.05, 0.20)$, and a per-point relative-error bound of 15%. The machine-readable record is distributed with the code. All four values are reported; there is no coupling fit.

Table 1: Preregistered comparison. Rates are in units of J . At zero reference, 0.000% denotes exact equality with zero absolute error, as recorded in the baseline log; no division by zero is performed. The finite- ϵ control is evolved from the same states under H_0 .

ϵ	R_ϕ (statevector)	R_ϕ (PDE)	Rel. error	Linear-limit observable	Linear-limit check	Notes
0.00	0.000000	0.000000	0.000%	18/18 paired modes; max. error 4.20%	pass (< 5%)	$H(0) = H_0$ exactly
0.10	0.030007	0.030000	0.024%	$H(0) = H_0$ exactly	pass (18/18; 4.20%)	$R_\phi(H_0) = 0$; full 2^{12} ; NLSE pass
0.20	0.060010	0.060000	0.017%	$H(0) = H_0$ exactly	pass (18/18; 4.20%)	$R_\phi(H_0) = 0$; full 2^{12} ; NLSE pass
0.40	0.119989	0.120000	0.009%	$H(0) = H_0$ exactly	pass (18/18; 4.20%)	$R_\phi(H_0) = 0$; full 2^{12} ; NLSE pass
0.80	0.239725	0.240000	0.115%	$H(0) = H_0$ exactly	pass (18/18; 4.20%)	$R_\phi(H_0) = 0$; full 2^{12} ; NLSE pass

Independent PDE calculation. We integrated Eq. (8) on 256 periodic points with a Fourier split-step method [Feit et al., 1982], step $10^{-4}J^{-1}$, through $Jt = 0.2$. For each ϵ and density, an ordinary least-squares phase slope over $t = 0, 0.02, \dots, 0.20$ determines Ω . These reference files were finalized before statevector residuals were evaluated. The split-step and analytic homogeneous rates agree to at worst 1.83×10^{-13} relative error.

Full-space emulation. The nonlinear runs use $N = 12$, $J = \hbar = a = 1$, $\Delta_j = 0$, and the dense 4096-component initial vector

$$|\Psi_\rho(0)\rangle = \bigotimes_{j=1}^{12} (\sqrt{1-\rho}|0\rangle_j + \sqrt{\rho}|1\rangle_j). \quad (11)$$

No excitation-sector block is ever constructed. A first-order product formula applies even bonds, odd bonds, and then the commuting $Z + ZZ$ layer, with $\delta J = 0.02$ [Suzuki, 1990]. Every trajectory stores all 11 full statevectors. The observable $B_{\text{bulk}} = (N-2)^{-1} \sum_{j=2}^{N-1} \langle \sigma_j^+ \rangle$ excludes the coordination-one endpoints. Its unwrapped phase is fit over the same samples as the PDE. At every finite coupling we separately rerun the two initial states under H_0 and form Eq. (10).

The high-density state places 72.512% of its probability in sectors with two or more excitations (the single-excitation weight is 20.616%). This weight is present at initialization and is conserved to 7.2×10^{-16} , because both H_0 and ZZ commute with total occupation. Consequently, the model cannot transfer probability between number sectors; its nonlinearity instead arises from interaction-dependent relative phases and entanglement among amplitudes already outside the single-excitation sector. A claimed inter-sector transfer under this Hamiltonian would violate its exact $U(1)$ symmetry.

Linear reproduction. At exactly $\epsilon = 0$ we use $N = 9$ and the original $\delta J = 0.8$. For $\Delta/J = 0$ and $1/4$, each exact one-excitation eigenmode is paired with the vacuum and its frequency is extracted from the phase of the projected $\langle X_j + iY_j \rangle$ signal over 12 steps. We also launch the normalized two-site packet $(|1_1\rangle + |1_2\rangle)/\sqrt{2}$ through vacuum, a sharp $\Delta/J = 1$ region beginning at zero-based site 5, and $\Delta_j/J = \exp[-(j-5)^2/2]$. A second call through the augmented code path at $\epsilon = 0$ provides the implementation-identity check.

Dynamic multi-excitation witness. In an additional full-space diagnostic we initialize the computational-basis state with adjacent excitations on zero-based sites 5 and 6, set $\epsilon = 0.8$, and evolve through $Jt = 2.0$ with the same $\delta J = 0.02$. We measure population transferred from the initially occupied basis configuration to every other configuration with at least two excitations, the adjacent-pair population, and the total-variation (TV) distance between the complete ZZ and matched- H_0 basis-population distributions. The calculation allocates the full 4096-component vector; number-sector weights are diagnostics only. Sparse continuous-time evolution supplies an independent cross-check at the final time.

5 Results

5.1 Parameter-free NLSE agreement

Table 1 and Fig. 2 give the complete registered sweep. The statevector rates are 0.030007, 0.060010, 0.119989, and 0.239725, whereas the independently integrated NLSE gives 0.030000, 0.060000, 0.120000, and 0.240000. The largest relative error is 0.115%, more than two orders of magnitude

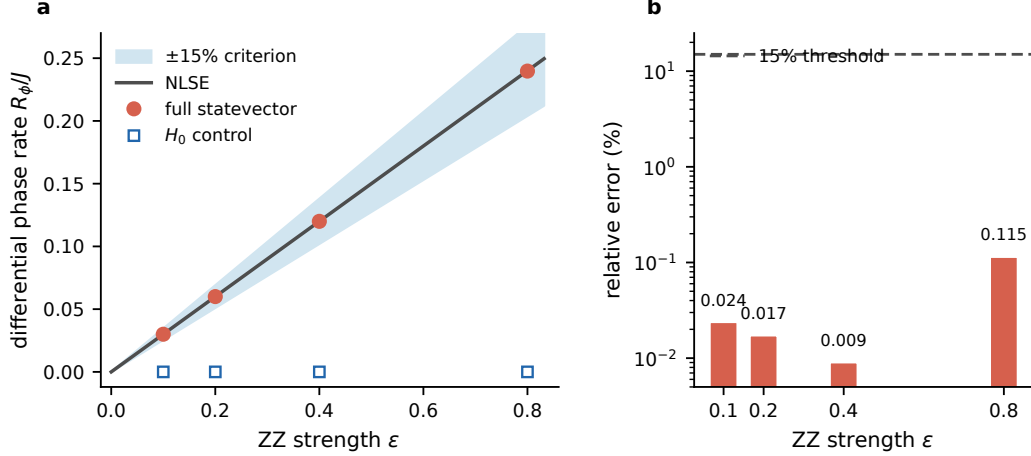


Figure 2: Nonlinear coefficient across the fixed eightfold coupling range. (a) Full-space statevector rates lie on the independently computed NLSE prediction; the unextended control is zero. The shaded band is the declared $\pm 15\%$ tolerance. (b) Pointwise errors on a logarithmic scale; labels give percentages.

below the 15% bound. The state norm drift is at most 2.0×10^{-15} . At $\epsilon = 0.8$, sparse continuous-time evolution gives $R_\phi/J = 0.239741$ (0.108% from the NLSE), showing that the agreement is not created by the product formula.

The near-zero residual has a specific explanation. A product state makes the intersite factorization exact at $t = 0$, and a homogeneous envelope annuls the hard-core gradient term in Eq. (6). The short fit estimates this local coefficient before substantial correlations form. Agreement is therefore a stringent check of the coefficient and simulator, but it is not evidence that the classical closure remains exact for all times.

The numeric zero-coupling row uses an exact-match convention necessitated by the zero PDE denominator. Separate statevector and split-step runs both give $R_\phi = 0.000000J$ and zero absolute error, so the displayed error is 0.000%; no division by zero is evaluated. The convention and both raw signals are retained in the baseline log.

5.2 The nonlinear signal is caused by ZZ

Figure 3 resolves the strongest-coupling run. The two intensities acquire different ZZ phases, their difference follows the NLSE slope, and both H_0 controls remain on zero phase. The full state becomes entangled: at $Jt = 0.2$ and $\rho = 0.20$, the half-chain entropy is 0.007864 bits under H versus 2.52×10^{-6} bits under H_0 . Thus the result is not obtained by evolving only a classical amplitude or a one-excitation vector. It is extracted from a multi-sector quantum state whose early-time first moment has a classical envelope limit.

5.3 Population transport beyond one excitation

Figure 4 supplies a direct dynamical witness that the emulator does not merely carry inert initial weight outside the single-excitation subspace. At $\epsilon = 0.8$ and $Jt = 2.0$, 74.132% of the initial adjacent-pair population has moved into 60 other two-excitation basis configurations above probability 10^{-8} . All of this transferred population lies outside the single-excitation subspace. The ZZ population distribution has 11.220% TV distance from the matched H_0 distribution; sparse exact evolution gives 11.221%. The attractive interaction retains 38.430% population on adjacent pairs, compared with 28.098% under H_0 .

This is population redistribution within the multi-excitation manifold, not creation of particles. The two-excitation probability remains 0.999999999999992 at the final sample and inter-sector transfer is exactly zero by $U(1)$ symmetry. Thus finite ZZ changes measured populations via interaction-

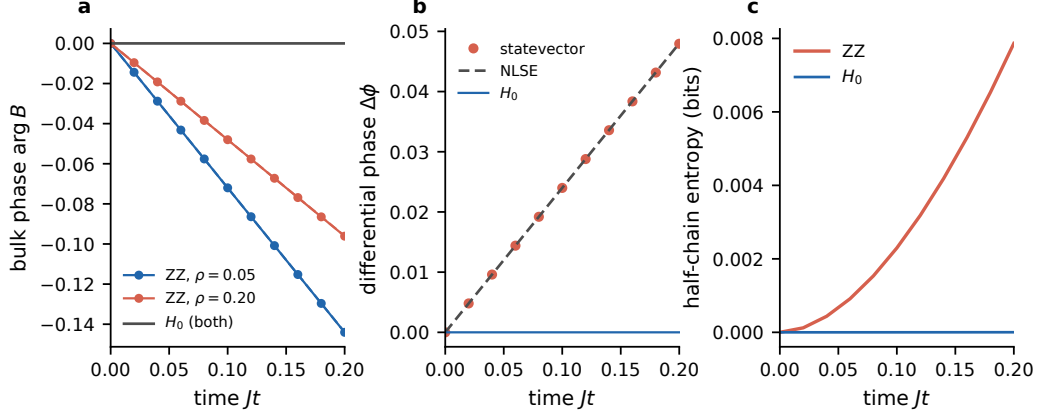


Figure 3: Mechanism control at $\epsilon = 0.8$. (a) Bulk phases for the two densities; dashed curves are independent NLSE solutions. The H_0 phases overlap at zero. (b) The density-differential phase has slope 0.239725, while the H_0 difference is zero. (c) Half-chain entropy confirms interacting full-space dynamics; it is not used in the phase-rate fit.

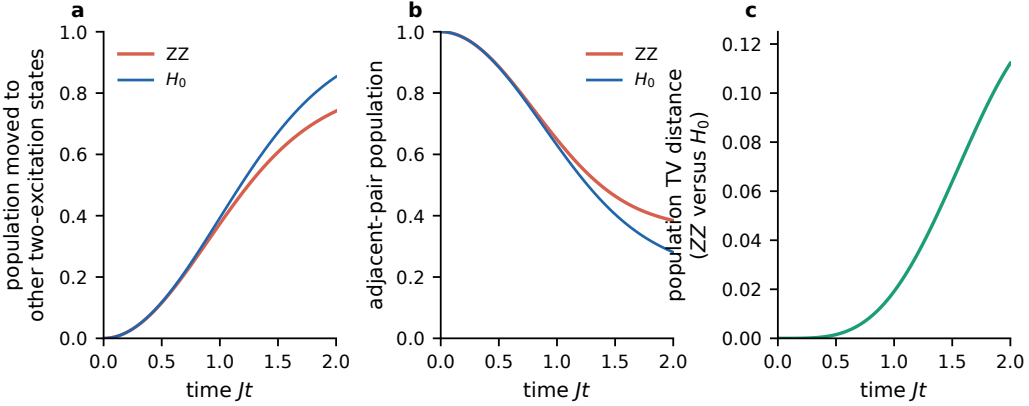


Figure 4: Dynamic full-space witness at $\epsilon = 0.8$ from an adjacent two-excitation basis state. (a) Population transferred to other two-excitation configurations. (b) Attractive ZZ retains greater adjacent-pair population than H_0 . (c) TV distance between the full computational-basis population distributions. The state remains in the two-excitation sector, but no sector projection is used.

dependent phase accumulation followed by hopping, while the number-conservation statement remains intact.

5.4 Exact linear reduction

The augmented code path and H_0 code path produce a maximum statevector difference of exactly zero at $\epsilon = 0$ for both coherent-state densities and all three propagation profiles. The side-by-side dispersion run independently evolves both implementations for every Ramsey mode and mass: across 18 trajectory pairs of shape 13×512 , the maximum statevector, projected-signal, and fitted-frequency differences are respectively 0, 0, and 0. Algebraically this follows immediately from Eq. (2); numerically it also rules out a hidden offset in the diagonal gate or Ramsey estimator.

Figure 5a overlays both Ramsey-extracted frequency sets against the exact open-chain eigenvalues: outer rings are augmented $H(0)$ results and inner dots are the separately evolved H_0 results. Every one of the 18 paired modes passes the 5% condition; the maximum error is 4.20%, and the zero mode has absolute error $1.3 \times 10^{-16}J$. The analytic open-chain formula and direct matrix diagonalization differ by at most $5.6 \times 10^{-16}J$. Figure 5b reproduces the qualitative wave-packet controls. The

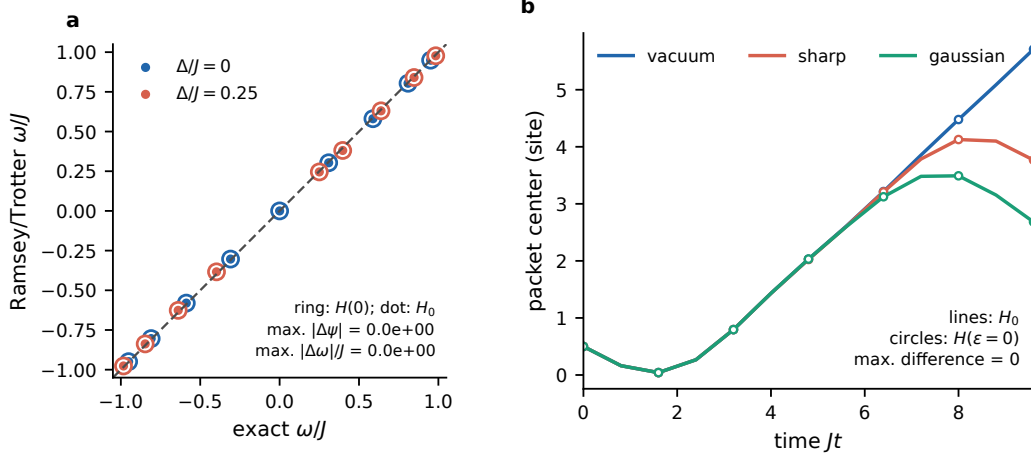


Figure 5: Linear-limit validation. (a) Side-by-side Ramsey/Trotter frequencies at the original step $\delta J = 0.8$ versus exact eigenvalues. Outer rings show augmented $H(0)$ and concentric inner dots show separately evolved H_0 ; no visual offset is applied. (b) Packet centers in vacuum and for sharp and Gaussian overdense profiles. Lines show H_0 ; open circles show the augmented implementation evaluated at exactly $\epsilon = 0$.

Table 2: Median wall time for 200 local-gate steps (seven runs); memory is for 201 stored complex 128 states.

N	2^N	Time (s)	Stored states (MiB)
9	512	0.0401	1.57
10	1024	0.0532	3.14
11	2048	0.0797	6.28
12	4096	0.1272	12.56

vacuum packet reaches center 5.710 at $Jt = 9.6$. The sharp and Gaussian barriers turn around at $Jt = 8.0$, with centers decreasing from 4.127 to 3.767 and from 3.491 to 2.683, respectively. Open symbols are the augmented $\epsilon = 0$ results and coincide exactly with H_0 .

5.5 Convergence and breakdown

Changing δJ from 0.08 to 0.01 at $\epsilon = 0.8$ keeps the NLSE error between 0.092% and 0.124%; the exact evolution error is 0.108% (Fig. 6b). The nonmonotonic ordering is small cancellation between product-formula and mean-field errors, not failed convergence.

The classical closure does fail on longer windows. Keeping every physical and numerical parameter fixed, the error grows from 0.115% at $Jt = 0.2$ to 8.04% at 1.6 and 13.64% at 2.0. The first post-confirmatory failure is $Jt = 2.4$, where the error is 21.4%; by $Jt = 4.0$ it is 70.27%. Meanwhile the high-density half-chain entropy rises from 0.007864 bits at 0.2 to 0.4460 bits at 2.4 and 0.8749 bits at 4.0. Correlation growth, hard-core saturation, and finite boundaries are all absent from the cubic factorization, so this trend is the expected failure mode rather than a numerical instability.

6 Computational cost and scope for quantum simulation

Applying a local two-qubit layer to a dense state costs $\mathcal{O}(N2^N)$ time and $\mathcal{O}(2^N)$ working memory per step. Table 2 reports direct seven-repeat benchmarks; 200 steps on 12 qubits take 0.127 s median, and 201 uncompressed stored states occupy 12.56 MiB. Over only $N = 9$ to 12 the observed time grows by a factor of 3.17, less than its asymptotic exponential because fixed Python/array overheads remain important. The 256-point, 2000-step PDE benchmark takes 0.051 s median and scales as $\mathcal{O}(M \log M)$ per split step.

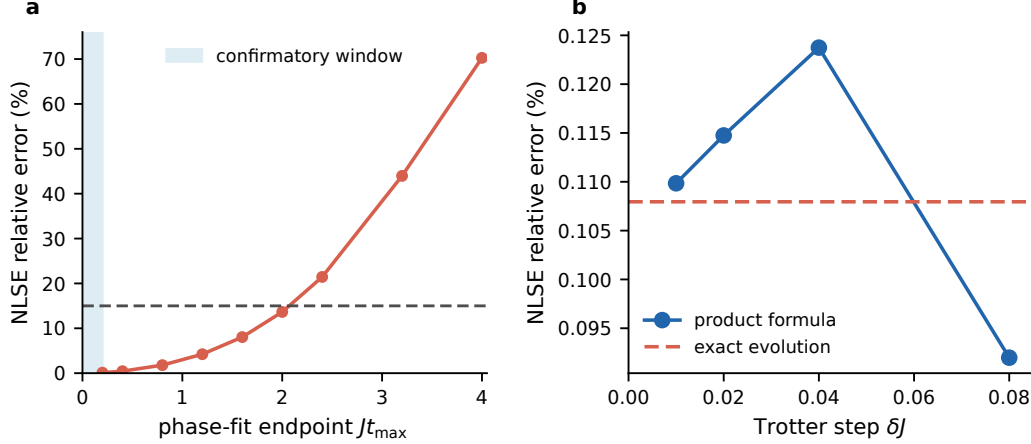


Figure 6: Validity boundary. (a) Extending the phase-fit endpoint at fixed $\epsilon = 0.8$ crosses the 15% line at $Jt = 2.4$; the shaded interval is the fixed validation window. (b) Short-window agreement is stable under step refinement and sparse exact evolution.

This small benchmark does not demonstrate quantum advantage. Indeed, with $\Delta = 0$ the uniform nearest-neighbor chain is gauge-equivalent to a homogeneous XXZ chain and retains special integrable structure. Nonuniform Δ_j , higher-dimensional geometries, general initial states, and local observables still produce interacting many-body dynamics for which a generic dense representation is exponential; tensor networks may postpone but not remove entanglement growth. A quantum processor implements each product step with $\mathcal{O}(N)$ local gates and stores the 2^N amplitudes implicitly [Lloyd, 1996]. Establishing an end-to-end advantage would additionally require state preparation, fault-tolerant depth, and measurement cost, none of which is benchmarked here. This positioning differs from amplitude-encoded linear solvers and from quantum simulations of microscopic stopping power [Rubin et al., 2024].

7 Limitations

The controlled result is narrower than a general nonlinear plasma solver. First, homogeneous SPM removes spatial derivatives, so the experiment validates q but not nonlinear packet transport, sideband growth, or collapse. Second, $N = 12$ leaves only ten coordination-two sites; boundary and lattice corrections become visible at long times. Third, the spin-1/2 saturation term in Eq. (6) is not present in a weakly occupied bosonic field and contributes when gradients develop. Fourth, the coherent initial state has substantial number fluctuations. Its multi-excitation support is essential, yet total number is exactly conserved, so this model cannot describe particle-production channels. Fifth, collisionless damping, kinetic trapping, ion response, dissipation, and the plasma-specific calibration of ϵ are outside the cubic local model. Finally, all runs are noiseless classical emulations; a hardware study would require calibrating longitudinal phases and mitigating noise without rescaling away the nonlinear shift.

These restrictions identify concrete next tests: a gapped carrier with resolved band curvature, a longer chain supporting several modulation wavenumbers, comparison of sideband growth with the focusing-NLSE spectrum, and an inhomogeneous coefficient calibrated to a specified plasma closure. They should be undertaken without changing the K_{ij} model after seeing the data.

8 Conclusion

A nearest-neighbor attractive ZZ interaction converts the linear plasma-wave qubit chain into a local density-interacting model. Its exact density identity fixes a cubic envelope coefficient $q = -2\epsilon J$. Full- 2^{12} statevectors reproduce the resulting intensity-differential phase rate across an eightfold coupling sweep with at most 0.115% error, while H_0 gives no effect and the augmented model

recovers H_0 exactly at zero coupling. Separately, finite ZZ produces a 11.220% change in the population distribution of an evolving two-excitation state, directly exposing dynamics beyond the one-excitation subspace. The same data show where the correspondence ends: quantum correlations drive the long-window result beyond the classical tolerance. The central outcome is therefore both positive and bounded—a local ZZ term realizes the early-time SPM coefficient of a plasma NLSE, but it is not by itself a universal nonlinear plasma model.

Reproducibility statement

Source, immutable method records, symbolic algebra, full complex statevectors, independent PDE trajectories, raw phase fits, paired $H_0/H(0)$ linear-mode signals and statevectors, two-excitation population trajectories, timing repeats, and figure-generation code accompany the paper. Every displayed number is generated from a JSON, CSV, or NPZ record; no reference trajectory is embedded as a numerical literal in the simulator. Appendix C maps claims to artifacts.

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A Algebraic mapping details

For completeness, expanding one bond gives

$$-\frac{\epsilon J}{4} Z_j Z_{j+1} = -\frac{\epsilon J}{4} (1 - 2n_j)(1 - 2n_{j+1}) \quad (12)$$

$$= -\epsilon J n_j n_{j+1} + \frac{\epsilon J}{2} (n_j + n_{j+1}) - \frac{\epsilon J}{4}. \quad (13)$$

The symbolic residual between the two expressions is exactly zero, and the $\epsilon \rightarrow 0$ limit is exactly zero. For an interior site,

$$[a_j, H_{ZZ}] = \frac{\epsilon J}{2} a_j (Z_{j-1} + Z_{j+1}), \quad (14)$$

while

$$[a_j, H_0] = \frac{iJ}{2} Z_j (a_{j+1} - a_{j-1}) - \Delta_j (-1)^j a_j. \quad (15)$$

Taking product-state expectations yields Eq. (6). Neighbor expansion in the ZZ contribution is

$$\frac{\epsilon J}{2} (z_{j-1} + z_{j+1}) \alpha_j = \epsilon J [1 - (\rho_{j-1} + \rho_{j+1})] \alpha_j \quad (16)$$

$$= \epsilon J (1 - 2\rho) \alpha - \epsilon J a^2 (\partial_X^2 \rho) \alpha + \mathcal{O}(a^4), \quad (17)$$

which explicitly identifies the local cubic term and its leading lattice correction.

B Complete linear validation

For odd $N = 9$, hard-wall wave numbers are $ka = p\pi/(N + 1)$, $p = 1, \dots, 4$, and the paired branches obey

$$\omega_{p,\pm} = \pm \sqrt{\Delta^2 + J^2 \sin^2(p\pi/10)}. \quad (18)$$

The remaining unpaired mode is $\omega = \Delta$ in our basis convention. Table 3 lists both separately evolved values for every Ramsey mode, not only the extrema. The complete full-state trajectories and projected complex signals, rather than only their fitted frequencies, are retained for the equality check.

C Artifact-to-claim map

Table 3: All paired N=9 Ramsey/Trotter dispersion values. The separately evolved H_0 and augmented $H(0)$ frequencies coincide exactly. Nonzero-mode entries are relative errors against the exact eigenvalue. For the zero mode, the numeric entry is $100|\omega_{\text{fit}} - \omega_{\text{exact}}|/J$, i.e., the absolute error normalized by the fixed energy scale.

Δ/J	Exact ω/J	$H_0 \omega/J$	$H(0) \omega/J$	Error (%)
0.00	-0.951057	-0.949433	-0.949433	0.17
0.00	-0.809017	-0.804073	-0.804073	0.61
0.00	-0.587785	-0.580677	-0.580677	1.21
0.00	-0.309017	-0.303320	-0.303320	1.84
0.00	-0.000000	+0.000000	+0.000000	1.32×10^{-14}
0.00	+0.309017	+0.303320	+0.303320	1.84
0.00	+0.587785	+0.580677	+0.580677	1.21
0.00	+0.809017	+0.804073	+0.804073	0.61
0.00	+0.951057	+0.949433	+0.949433	0.17
0.25	-0.983366	-0.976661	-0.976661	0.68
0.25	-0.846764	-0.837660	-0.837660	1.08
0.25	-0.638742	-0.626264	-0.626264	1.95
0.25	-0.397481	-0.383725	-0.383725	3.46
0.25	+0.250000	+0.245616	+0.245616	1.75
0.25	+0.397481	+0.380783	+0.380783	4.20
0.25	+0.638742	+0.630495	+0.630495	1.29
0.25	+0.846764	+0.840312	+0.840312	0.76
0.25	+0.983366	+0.977300	+0.977300	0.62

Table 4: Packet-propagation diagnostics at $\epsilon = 0$. Sites are zero based.

Profile	initial CoM	peak CoM	Jt_{peak}	final CoM
Vacuum	0.500	5.710	9.6	5.710
Sharp	0.500	4.127	8.0	3.767
Gaussian	0.500	3.491	8.0	2.683

Table 5: Primary audit artifacts, relative to the submission directory.

Artifact	Contents
code/method_plan.md, json, py	Model, PDE, densities, sweep, tolerance, time window, and timestamps fixed before runs.
logs/mapping_symbolic.json	SymPy density identity, zero-coupling limit, μ_ϵ , and q .
logs/pde_reference_precomputed.json	Independently computed analytic and split-step rates, solver grid, time step, source hash, and wall time.
logs/nonlinear_summary.json	All four residuals, raw phase fits, norms, sector weights, entropy, H_0 controls, exact check, and hashes.
logs/nonlinear_eps*.npz	Eleven full 4096-component statevectors for every coupling, density, and model control.
logs/linear_summary.json	Both sets of all 18 frequencies, trajectory-level dispersion identity metrics, propagation diagnostics, exact- $\epsilon = 0$ comparisons, and artifact hashes.
logs/dispersion*.npz, propagation*.npz	Paired H_0 and $H(0)$ full dispersion statevectors and complex Ramsey signals; complete packet statevectors and densities.
logs/additional_checks.json	Step convergence, sparse-exponential cross-check, long-time failure, entropies, and timing repeats.
logs/full_hilbert_population_transfer.json, multiexcitation_transfer_eps0.80.npz	Dynamic population transport among two-excitation configurations, ZZ versus H_0 TV distance, exact cross-check, full statevectors, norms, and sector conservation.
logs/zero_baseline_headline.json	Numeric zero-coupling statevector and PDE rates, absolute error, and exact-match display convention.
logs/artifact_manifest.json	SHA-256 and byte count for generated figures and table fragments.

D Numerical definitions and sanity checks

Qubit zero is the least-significant computational-basis bit. Each two-qubit gate gathers the four amplitudes (00, 01, 10, 11) for its target pair, applies the exact 4×4 exponential of the local bond, and scatters them back. The diagonal $Z + ZZ$ layer is applied as a basis-dependent phase. Sparse Hamiltonians are constructed independently from basis transitions for the continuous-time cross-check. Across confirmatory nonlinear trajectories the largest norm error is 2.0×10^{-15} ; across the linear trajectories it is 2.2×10^{-15} . Excitation-sector probabilities are computed only as diagnostics after evolution, never used to reduce the state.

The initial high-density binomial sector weights are $P_1 = 0.2061584302$ and $P_{\geq 2} = 0.7251220931$; for the low-density state they are $P_1 = 0.3412800554$ and $P_{\geq 2} = 0.1183598570$. At finite ϵ the maximum drift of any sector weight is 7.2×10^{-16} . These values directly test both use of the full Hilbert space and exact number conservation.

For the paired dispersion identity check, each mass-setting file stores one $9 \times 13 \times 512$ full-state array for H_0 and another for augmented $H(0)$, together with both projected complex-signal arrays. Across all 18 mode pairs, the maximum statevector, signal, and fitted-frequency absolute differences are 0, 0, and 0, respectively.

For the separate adjacent-pair witness, the initial state is entirely in the two-excitation manifold. By $Jt = 2.0$, 74.132% has transferred from the initial basis configuration to other multi-excitation configurations and the finite- ZZ versus H_0 population TV distance is 11.220%. The final two-excitation weight is 0.999999999999992. This distinguishes nontrivial population dynamics outside the one-excitation subspace from the forbidden claim of transfer between number sectors.